

A unipolar-driven synaptic transistor for environment-adaptable vision system

Received: 24 March 2025

Accepted: 10 August 2025

Published online: 16 August 2025



Sukwon Jang^{1,5}, Keunho Soh^{2,5}, Chungryeol Lee^{1,5}, Taehyun Nam¹, Minjae Jang¹, Jeong-ik Park¹, Changhyeon Lee¹, Junhwan Choi³, Jung Ho Yoon² ✉ & Sung Gap Im^{1,4} ✉

Unlike conventional synapses in neural networks relying on bipolar spike-driven modulation, biological synapses in the peripheral nervous system handle unipolar input based on stimulus intensity, generating excitatory or inhibitory signals. Here, we demonstrate a unipolar-driven synaptic transistor (UDST) that exhibits both responses under a single-polarity voltage. The unipolar property is achieved within a single device through a bilayer gate dielectric, comprising a high- k charge trapping layer ($k > 6$) and an ultrathin charge tunneling layer (< 5 nm), which synergistically facilitate dipole polarization and charge trapping. The UDST exhibits potentiation, depression, and adaptation while maintaining exceptional durability, with a dynamic range reduction of less than 0.9% over 2000 potentiation-depression cycles and minimal conductance variation of only 0.3%. This work presents the first implementation of a self-adaptive artificial vision system based on a unipolar-driven synaptic device, utilizing a 3×3 UDST array to achieve real-time object tracking and adaptive sensory processing without external control or computational peripheral circuits.

Artificial sensory systems that emulate human sensory functions are crucial for the development of smart healthcare systems, human-machine interfaces, and prosthetics^{1–4}. The human sensory system includes the sensory receptors for external stimuli detection and afferent neurons for the signal transduction to the brain to process and interpret the sensory information⁵. Synapses between sensory receptors and afferent neurons facilitate parallel information processing with high energy efficiency. They enable sensory systems to locally amplify, inhibit or adapt signals through neuromodulation of excitatory and inhibitory synapses, thereby regulating the action potential transmission and neuronal responses to incoming stimuli⁶. Accordingly, recent advances in neuromorphic devices have enabled the development of artificial sensory nerves and reflex arcs that emulate such biological processes. In particular, significant progress has been made in closely mimicking biological functions through the

integration of sensors, preprocessing circuits, and synaptic devices^{7,8}. Furthermore, the interaction between synaptic device and peripheral actuators⁹ as well as the integration of different types of sensory inputs had also been explored intensively¹⁰.

Among sensory systems, the visual adaptation of the retina exemplifies how biological systems adjust to enable vision across a wide range of illumination intensities, offering valuable inspiration for the development of artificial vision systems capable of dynamic light adaptation^{11–15}. In the human visual system, the optical signal is captured and converted into electrical potential by photoreceptors, specifically cones and rods, in the retina¹⁶. Subsequently, the input electrical potential is processed by horizontal cells and bipolar cells, which form different neural circuits to pre-process the visual information (Fig. 1a)¹⁷. The visual adaptation, including both scotopic (dim to bright light) and photopic (bright to dim light) adaptation, is

¹Department of Chemical and Biomolecular Engineering Korea Advanced Institute of Science and Technology (KAIST), Yuseong-gu, Korea. ²School of Advanced Materials Science and Engineering Sungkyunkwan university, Jangan-gu, Suwon, Republic of Korea. ³Department of Chemical Engineering Dankook University, Suji-gu, Yongin, Republic of Korea. ⁴KAIST Graduate School of Semiconductor Technology, KAIST, Yuseong-gu, Korea. ⁵These authors contributed equally: Sukwon Jang, Keunho Soh, Chungryeol Lee. ✉e-mail: junghoyoon@skku.edu; sgim@kaist.ac.kr

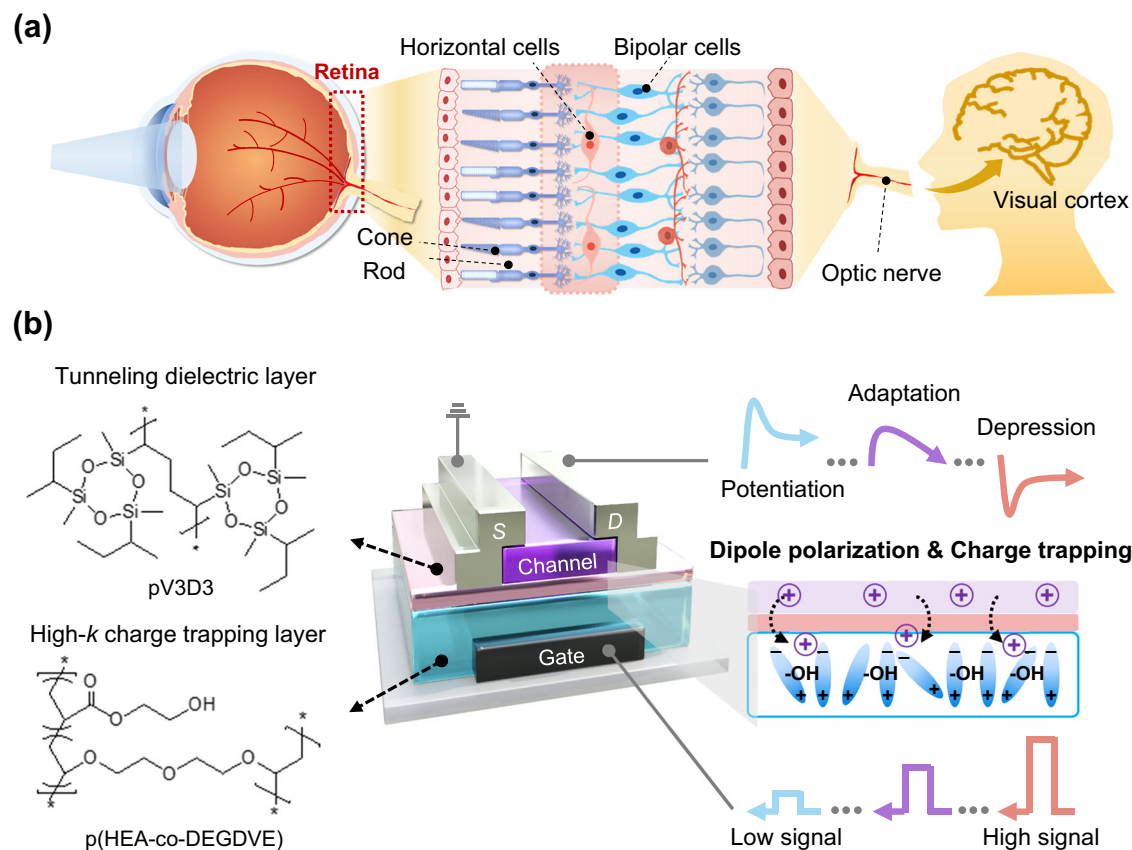


Fig. 1 | Visual adaptation of the retina. a A schematic illustration of the neural circuits (photoreceptor-bipolar cell-horizontal cell) in retina. **b** Design of the UDST and its three dynamic reconfigurable operating states (potentiation, adaptation, depression) governed by the magnitude of the presynaptic voltage pulses.

primarily regulated by a gradual shift in dominance between rod cells and cone cells¹⁸. Rod cells predominantly mediate scotopic vision, while cone cells are active in photopic vision. These photoreceptors adjust the visual threshold, determining the minimum illuminance required to perceive visual stimuli, thereby facilitating photopic and scotopic adaptation over a wide range of illumination conditions¹⁹. Here, visual thresholds for scotopic and photopic adaptation are influenced by synaptic dynamics between photoreceptors and afferent neurons²⁰. These dynamics are specifically mediated by feedback mechanisms from bipolar cells and horizontal cells, which respectively generate excitatory postsynaptic current (EPSC) and inhibitory postsynaptic currents (IPSC)²¹. These complementary yet opposing functions regulate potentiation, increasing photoreceptor sensitivity under dim lighting, and depression, decreasing sensitivity for adaptation to bright illumination, thereby enabling vision across a wide range of light intensities.

While significant progress has been made in developing artificial synapses, most research has focused on implementing synaptic behaviors for artificial neural networks or spiking neural networks, which are inspired by synaptic transmission in the central nervous system (CNS)^{22,23}. In such systems, synaptic weight modulation is driven by pre- and post-neuron spikes, typically requiring bipolar signals to enable both potentiation and depression. However, this design differs fundamentally from unipolar dynamics observed in biological synapses of the peripheral nervous system (PNS) that preprocess external stimuli before transmitting signals to the brain, where external stimuli inherently follow unipolar behavior and exhibit intensity-dependent variations²⁴.

To bridge this gap, the development of unipolar-driven synapses offers a biologically inspired approach to sensory processing by directly reflecting the unipolar dynamics of the PNS. Unlike

conventional artificial synapses, unipolar-driven synapses exhibit both excitatory and inhibitory synaptic responses solely by the magnitude of voltage⁷. This novel approach eliminates the need for polarity inversion or additional biasing elements, enabling a simplified yet highly efficient sensory processing architecture. Additionally, unipolar-driven synapses enable adaptive synaptic plasticity, allowing real-time preprocessing of sensory inputs by dynamically modulating the synaptic responses based on stimulus intensity. By pioneering an artificial counterpart to biological synapses of PNS, unipolar-driven synapses establish a new paradigm for self-regulating artificial sensory systems.

Despite such benefits, achieving the unipolar-driven synaptic devices still remains highly challenging, as inducing opposing effects of potentiation and depression under the same voltage polarity usually conflicts with the conventional mechanisms of charge transport²⁵. Although several recent reports explored approaches such as ferroelectric polarization or electrochemical doping to overcome the conflict, limitations still persist. These include a significant imbalance in conductance changes between potentiation and depression²⁶, as well as increased device complexity due to the need for additional input terminals to realize unipolar synapse operation²⁷. Due to such difficulties in implementing unipolar synapses, most synaptic devices still rely on bipolar electrical signals to mimic synaptic potentiation and depression. As a result, these systems often require external load gates or bias control circuits to approximate adaptive synaptic plasticity, rather than intrinsically processing the intensity-dependent sensory inputs as observed in biological synapses of PNS²⁸.

In this study, we demonstrate a unipolar-driven synaptic transistor (UDST) by developing a bilayer gate dielectric composed of two polymer layers: a high-k charge trapping layer (HkCTL), made of poly(2-hydroxyethyl acrylate-co-diethylene glycol divinyl ether)

[p(HEA-co-DEGDVE)] and a tunneling dielectric layer (TDL), made of poly(1,3,5-trivinyl-1,3,5-trimethyl cyclotrisiloxane) (pV3D3) (Fig. 1b). The hydroxyl groups (-OH) in p(HEA-co-DEGDVE) HkCTL were capable of inducing both dipole polarization and effective charge trapping due to their strong polarity²⁹. Dipole polarization enhanced the gate electric field by aligning molecular dipoles, whereas charge trapping counteracted this effect by locally screening the field. This interplay dynamically modulated the gate electric field, enabling the UDST to exhibit three reconfigurable synaptic behaviors of potentiation, depression, and adaptation (potentiation-to-depression transition). Most importantly, these behaviors were driven by a single-polarity voltage input, where the magnitude of the input signal governed state transitions, mirroring the intensity-dependent synaptic modulation observed in the PNS. The degree of potentiation and depression was controlled systematically by adjusting the thickness of the p(HEA-co-DEGDVE) HkCTL and pV3D3 TDL, which enabled us to achieve the balanced potentiation and depression characteristics. Furthermore, the dynamic range of potentiation and depression could also be controlled effectively by adjusting the chemical composition of the p(HEA-co-DEGDVE) HkCTL. The tunable dynamic range facilitated the UDST to emulate the preprocessing functions of the PNS, where variations in receptor density and synaptic connectivity modulate the signal magnitude transmitted to the brain, even for the same stimuli. The UDST demonstrated exceptional durability with its dynamic range decrease by less than 0.9% over 2000 pulses of repetitive potentiation and depression cycles while maintaining a minimal conductance variation of only 0.3% between potentiation and depression. This marks the first demonstration of unipolar-driven artificial synapses achieving both well-balanced plasticity updates and cycle endurance. Moreover, by fabricating 3 by 3 UDST array integrated with a CdS sensor, an environment-adaptable artificial vision system was realized. In this system, the UDSTs served as adaptive processing units that dynamically adjusted their sensitivity to each illumination change by means of potentiation and depression. This capability allowed the output current of the synapses to be modulated not only by instantaneous light intensity but also by prior exposure condition, enabling a temporally adaptive response. Thus, even in the absence of complex external circuits or computational algorithms, the UDSTs could provide context-dependent sensory processing, making them highly advantageous for adaptive artificial sensory systems. In a practical scenario involving moving object, the system successfully demonstrated object tracking as well as photopic and scotopic adaptation without requiring computational peripheral circuits such as topological inverter or comparator, nor additional synapses beyond the UDST itself. To the best of our knowledge, this work represents the first implementation of a self-adaptive artificial vision system based on unipolar-driven synaptic device, mimicking PNS-like preprocessing of sensory inputs and enhancing image perception across diverse lighting environments.

Results

Synthesis and chemical characterization of the bilayer gate dielectric

HEA, an acrylic monomer containing polar hydroxyl group, -OH, was selected to facilitate the induction of dipole polarization and charge trapping effects in the UDST^{30,31}, while DEGDVE was copolymerized with HEA as a cross-linker to improve the dielectric strength of the copolymer dielectric layer^{32,33}. The p(HEA-co-DEGDVE) served as the primary gate dielectric in the UDST, blocking leakage current between the channel and the gate, while also functioning as the HkCTL that enabled the UDST to implement potentiation and depression through the dipole polarization or charge trapping effects in response to input voltage pulses. A siloxane-based polymer pV3D3 features a low dielectric constant ($k \sim 2.2$) as well as exceptional insulating properties, even at sub-10 nm-scale thickness³⁴. As a TDL, pV3D3 passivates

effectively the polar surface functionalities of the p(HEA-co-DEGDVE), and thus reducing the interfacial trap states between the semiconductor and the gate dielectric³⁵. The use of pV3D3 TDL not only prevents device degradation but also ensures a clear distinction between dipole polarization at low-voltage pulses and charge trapping effects at high-voltage pulses, allowing the UDST to demonstrate reliably both potentiation and depression characteristics in response to unipolar input. The p(HEA-co-DEGDVE) and pV3D3 were deposited using initiated chemical vapor deposition (iCVD), a vapor-phase polymer deposition process advantageous for precise film thickness control and well-defined multilayered structure^{36–39}. The -OH content in p(HEA-co-DEGDVE), correlated closely to the extent of dipole polarization and charge trapping effects, could be controlled precisely as intended by adjusting the composition of the copolymer through the modulation of HEA-to-DEGDVE input monomer ratios during the iCVD process. A series of pHEA homopolymer and p(HEA-co-DEGDVE) copolymers with various HEA-to-DEGDVE compositions were synthesized and designated as pHID2, pHID3, and pHID5 according to the input monomer ratios. Fourier-transform infrared (FT-IR) spectroscopy was employed to analyze the preservation of key chemical functionalities in the pV3D3, pHEA and p(HEA-co-DEGDVE) copolymers (Fig. 2a). The FT-IR spectra of pV3D3 exhibited all the characteristic peaks corresponding to the Si-C group (700–800 cm^{-1}) and the Si-O-Si group (1000–1100 cm^{-1}), confirming the presence of the structural motifs of the pV3D3³⁴. The FT-IR spectra also revealed all the characteristic peaks for the -OH group (3100–3600 cm^{-1}), the C=O group (1750 cm^{-1}), the O=C-O group (1160 cm^{-1}), and the C-O-C group (1090 cm^{-1}), confirming the successful synthesis of the pHEA homopolymer and p(HEA-co-DEGDVE) copolymer films⁴⁰. As the DEGDVE input flow increased, the peaks corresponding to the -OH and O=C-O groups associated with HEA diminished, while the C-O-C peak from DEGDVE intensified. High-resolution X-ray photoelectron spectroscopy (XPS) analysis elucidated the chemical composition of the pHEA homopolymer and p(HEA-co-DEGDVE) copolymers, further confirming these findings (Fig. 2b). In the XPS $C 1s$ spectra, the peak corresponding to the C^*-O bond (at 286.1 eV), prevalent in the DEGDVE cross-linker, increased with a higher DEGDVE input flow, while the peak corresponding to the C^*-O bond (at 288.5 eV), specific to the HEA monomer, decreased⁴¹. This trend matched well with the chemical composition of the copolymers as regulated by the input flow ratio. Together, the XPS and FT-IR results confirm the successful copolymerization of HEA and DEGDVE in p(HEA-co-DEGDVE), with their chemical composition precisely tunable through input flow ratio adjustment. To analyze the surface characteristics of pV3D3, pHEA, and p(HEA-co-DEGDVE) copolymers, atomic force microscopy (AFM) was performed (Fig. 2c). No apparent defects such as pinholes or dewetting were observed over the polymer films, with a root-mean-square roughness (R_q) of less than 1 nm. Furthermore, the AFM images showed no appreciable phase separation between HEA and DEGDVE, indicating the homogeneous copolymerization of p(HEA-co-DEGDVE). Their smooth and uniform morphology underscores the key advantage of the vapor-phase polymerization method, which is highly beneficial for ensuring reliable integration as a gate dielectric of transistors⁴². The dielectric properties of the pHEA homopolymer and p(HEA-co-DEGDVE) copolymers were evaluated using metal-insulator-metal (MIM) structure, in which polymer layer was sandwiched between top and bottom aluminum electrodes over an active area of $1 \times 1 \text{ mm}^2$. The dielectric constants of the pV3D3, pHEA and p(HEA-co-DEGDVE) copolymers were determined by analyzing the slope of charge capacitance per unit area (C_i) versus dielectric thickness (d) (Fig. 2d). A higher DEGDVE content in the p(HEA-co-DEGDVE) copolymers resulted in the decrease in the dielectric constant from 11.2 for pHEA to 5.3 for pHID5, reflecting the reduced content of highly polar -OH groups in the copolymers. The pV3D3 showed a low dielectric constant of 2.2, which was fully consistent with previous studies³⁴.

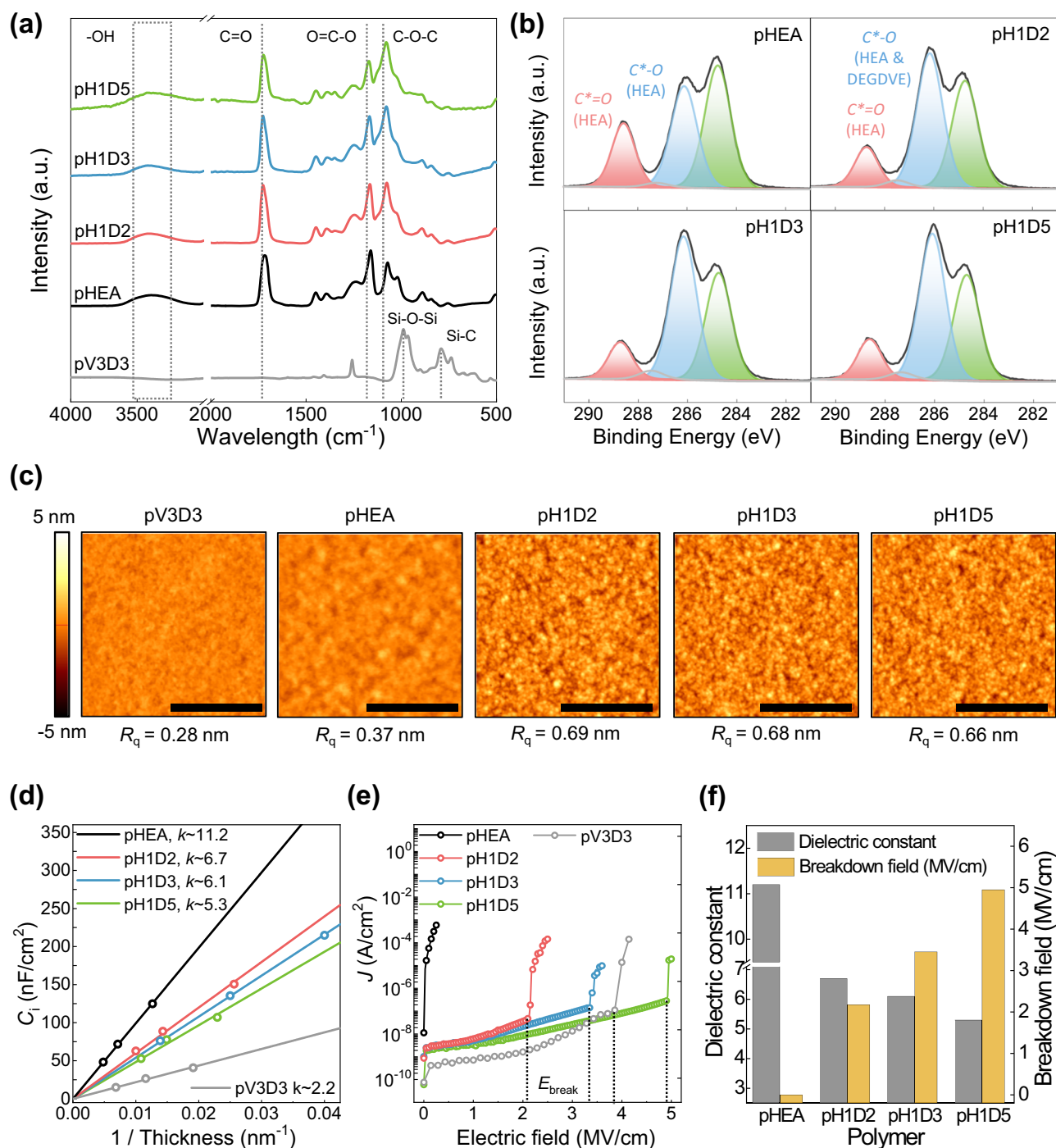


Fig. 2 | Synthesis and chemical characterization of the bilayer gate dielectric. **a** FT-IR spectroscopy of pV3D3, pHEA and p(HEA-co-DEGDVE) copolymers. **b** XPS C 1s spectra of pHEA and copolymers. **c** AFM images of pV3D3, pHEA and p(HEA-co-

DEGDVE) copolymers. Scale bar: 1 μ m. **d** The C_i vs $1/d$, **e** J vs E characteristics of pV3D3, pHEA and p(HEA-co-DEGDVE) copolymers. **f** Dielectric constant and E_{break} of pHEA with respect to the copolymer composition.

The insulating properties of the pV3D3, pHEA and p(HEA-co-DEGDVE) copolymers were assessed by measuring the breakdown field (E_{break}), defined as the electric field (E) at which the leakage current density (J) exhibited a sudden increase (Fig. 2e). The pV3D3 exhibited superior insulating properties with J below 10^{-9} A/cm² and a E_{break} greater than 4 MV/cm. Additionally, the pV3D3 layer represented the Fowler-Nordheim (F-N)-like tunneling behavior in the E range starting from approximately 2 MV/cm, which could facilitate the charge tunneling at higher E . Compared to the limited insulating performance of the pHEA homopolymer, the p(HEA-co-DEGDVE) copolymers exhibited

significant improvement. The incorporation of the DEGDVE cross-linker into the polymer matrix effectively enhanced the insulating properties^{43,44}, achieving J below 10^{-7} A/cm² and a E_{break} reaching 5 MV/cm for pH1D5. Notably, an increase in DEGDVE content within the copolymers led to a higher E_{break} ; however, this was accompanied by a reduction in the dielectric constant due to the enhanced cross-linked network (Fig. 2f). Given that pH1D3 demonstrated both a high dielectric constant ($k > 6$) and a reliable E_{break} (> 3 MV/cm), the pH1D3 was selected as the representative dielectric composition for the UDS.

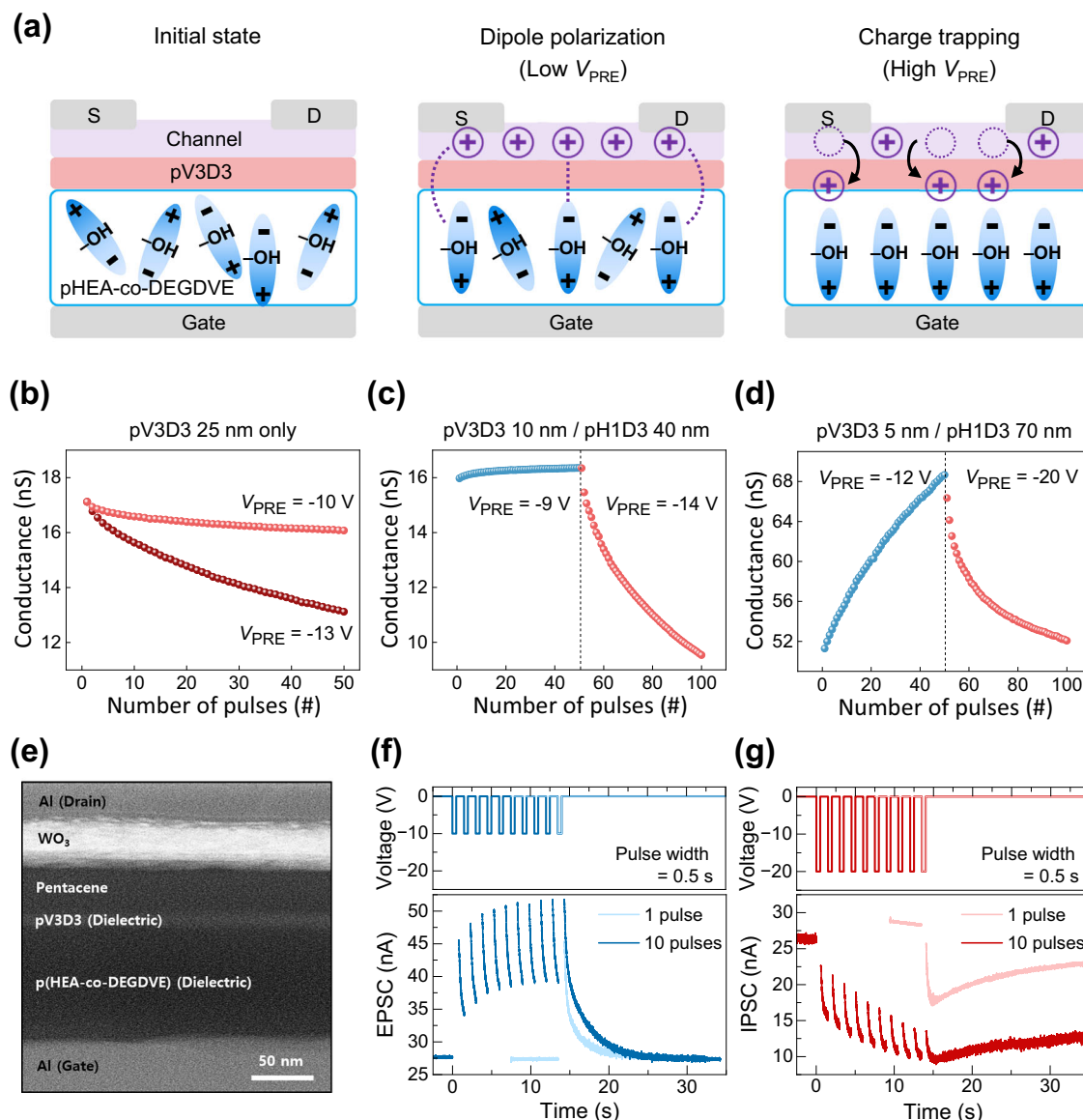


Fig. 3 | Operating principle and design of the UDST. **a** A schematic illustration of the operating mechanism of the UDST. **b** Synapse characteristics of the UDST with a 25 nm-thick pV3D3 TDL (UDST-1). **c** Synapse characteristics of the UDST with a 40 nm-thick pH1D3 HkCTL and a 10 nm-thick pV3D3 TDL (UDST-2). **d** Synapse characteristics of the UDST with a 70 nm-thick pH1D3 HkCTL and a 5 nm-thick pV3D3

TDL (UDST-3). **e** A cross-sectional high-resolution transmission electron microscopy (HRTEM) image of the UDST-3. **f** The spike-number-dependent plasticity (SNDP) for potentiation induced by dipole polarization and **(g)** by depression induced by charge trapping in the pH1D3 HkCTL.

Operating principle and design of the UDST

Figure 3a provides a schematic illustration of the operating mechanism of the UDST, depicting initial state, dipole polarization-induced hole accumulation at low negative presynaptic voltage pulses ($-V_{PRE}$), and charge trapping-induced hole depletion at high $-V_{PRE}$. At the initial state, where no electric field is applied, the -OH groups in the p(HEA-co-DEGDVE) HkCTL exhibit a randomized orientation. Upon application of a low $-V_{PRE}$, the -OH groups begin to align vertically along with the applied electric field, with the more electronegative oxygen atoms oriented toward the semiconductor. This alignment enhances the electric field toward the gate and induces hole accumulation at the semiconductor/dielectric interface, resulting in increased channel conductance (potentiation). When a higher $-V_{PRE}$ is applied, more -OH groups within the p(HEA-co-DEGDVE) HkCTL align toward the semiconductor. However, simultaneously, the strong electric field facilitates hole tunneling from the semiconductor through the pV3D3 TDL into the HkCTL, where the holes are subsequently trapped. Here, these

trapped holes substantially reduce the effective electric field toward the semiconductor, leading to hole depletion at the interface between the semiconductor and the pV3D3 TDL, accordingly leading to the reduced channel conductance (depression).

To support the proposed operation mechanism of the UDST, involving the competing effects of dipole polarization and charge trapping, the band alignments among pentacene, pV3D3, and pH1D3 were checked experimentally. The highest occupied molecular orbital (HOMO) levels were extracted using ultraviolet photoelectron spectroscopy (UPS), while the lowest unoccupied molecular orbital (LUMO) levels were estimated from reflection electron energy loss spectroscopy (REELS) for pH1D3, electron energy loss spectroscopy (EELS) for pV3D3, and UV-Vis spectroscopy for pentacene, respectively (Supplementary Figs. 1–4). The derived band diagram revealed a large HOMO offset between the pentacene channel (HOMO = 5.01 eV) and the pV3D3 tunneling dielectric layer (HOMO = 9.00 eV), corresponding to a substantial energy barrier of 4.99 eV for hole injection.

This barrier explains the necessity of a strong electric field to drive hole trapping in the pHID3 HkCTL. Consequently, the UDST enables both potentiation and depression under unipolar voltage pulses, with hole accumulation driven by dipole polarization of the -OH groups under a weak electric field (potentiation), and hole depletion occurring only under a strong electric field through hole injection (depression).

To validate the proposed operating mechanism, the synaptic characteristics of three different UDSTs, each with a distinct combination of pV3D3 TDL/pHID3 HkCTL layer thicknesses, were investigated by analyzing conductance changes in response to the applied V_{PRE} (Fig. 3b–d). Firstly, the UDST with a 25 nm-thick pV3D3 TDL without a pHID3 HkCTL (UDST-1), exhibited only a gradual decrease in channel conductance (depression) under a low V_{PRE} range of -10 V to -13 V, which was attributed to hole trapping within the pV3D3 with negligible contribution from the dipole polarization of the pV3D3 dielectric layer⁴⁵. In contrast, the UDST comprising a 40 nm-thick pHID3 HkCTL and a 10 nm-thick pV3D3 TDL (UDST-2) showed a slight increase in channel conductance (potentiation) at a low V_{PRE} of -9 V, suggesting that a weak electric field induced by dipole polarization of the -OH groups in the pHID3 HkCTL was transmitted to the channel, resulting in additional hole accumulation⁴⁶. Under a higher V_{PRE} of -14 V, UDST-2 exhibited a decrease in channel conductance (depression), attributed to the hole trapping in pHID3 HkCTL. Notably, UDST-2 demonstrated both potentiation and depression behavior without the need to change the polarity of V_{PRE} , though the dynamic range of potentiation and depression was skewed heavily toward depression, indicating an imbalance in synaptic characteristics. The UDST-3 was designed by increasing the thickness of the pHID3 HkCTL from 40 nm to 70 nm while reducing the thickness of the pV3D3 TDL from 10 nm to 5 nm from the UDST-2. This adjustment allowed the electric field generated by the applied V_{PRE} to be concentrated mostly on the thicker pHID3 HkCTL, enhancing the dipole polarization effect caused by the polar pHID3 HkCTL. As a result, UDST-3 showed a balanced increase in conductance (potentiation) under a low V_{PRE} of -12 V and a decrease in conductance (depression) under a high V_{PRE} of -20 V, with nearly identical dynamic range of potentiation and depression, establishing an optimal device configuration. These findings clearly demonstrate that incorporating both pV3D3 TDL and pHID3 HkCTL as a bilayer gate dielectric enables the emulation of opposite synaptic behaviors (potentiation and depression) under unipolar V_{PRE} , driven by the combined effects of dipole polarization and charge trapping within the pHID3 HkCTL. Notably, the degree of each potentiation and depression is systematically tunable by adjusting the thickness of the pV3D3 TDL and pHID3 HkCTL. This tunability implies that well-balanced synaptic behavior can be preserved even with scaled dielectric thickness, enabling low-voltage operation.

From the analysis above, UDST-3 was employed as the optimal device configuration for further electrical analysis and application to a self-adaptive artificial vision system. A cross-sectional high-resolution transmission electron microscopy (HRTEM) image of the UDST confirmed the precise and conformal deposition of the ultrathin pV3D3 (5 nm), pHID3 (70 nm) and WO₃ buffer layer (15 nm) on the Al gate, with the thicknesses accurately matching the design specifications (Fig. 3e). Furthermore, energy-dispersive X-ray (EDX) spectroscopy mapping verified the presence of discrete layer boundaries in the bilayer dielectric stack, with no evidence of intermixing between the layers (Supplementary Fig. 5). Such clearly defined bilayer of pV3D3 TDL/pHID3 HkCTL, as well as WO₃ buffer layer to reduce contact resistance at the electrodes/semiconductor interface⁴⁷, is crucial for the optimal unipolar operation and balanced potentiation/depression characteristics of the UDST.

The retention characteristics of the UDST were analyzed during both potentiation and depression. For this purpose, spike-number-dependent plasticity (SNDP) was investigated using a low V_{PRE} of -10 V with a 0.75 s for potentiation and high V_{PRE} of -20 V with a 0.5 s for

depression. The low V_{PRE} led to an increase in post-synaptic current (PSC), corresponding to hole accumulation at the semiconductor/dielectric interface (Fig. 3f). The EPSC rapidly decayed, returning to its initial current value within a few seconds. While the EPSC increased with the number of V_{PRE} , the improvement in retention was marginal, indicating short-term memory (STM) behavior for potentiation. On the other hand, the high V_{PRE} induced a decrease in PSC, reflecting hole depletion at the semiconductor/dielectric interface (Fig. 3g). The IPSC slowly approached its initial current level over tens of seconds but did not fully return, indicating long-term memory (LTM) behavior for depression. To substantiate the LTM capability, the UDST was programmed to five discrete conductance states using V_{PRE} of -22 V with a 0.05 s (Supplementary Fig. 6), which remained clearly distinguishable after 10,000 s with less than 11% variation, confirming stable multi-level retention. The distinct difference in retention behaviors between potentiation and depression supports the principle that the unipolar operation of the UDST is attributed to dipole polarization and charge trapping, respectively. The rapid decay of EPSC after low V_{PRE} aligns well with the STM characteristics of dipole polarization, where charges are temporarily accumulated and quickly dissipate once the field is removed. In contrast, the sustained IPSC after high V_{PRE} corresponds to LTM, as expected from charge trapping, where charges remain trapped for an extended period, resulting in slower recovery⁴⁸. Meanwhile, the UDST also performed fundamental functions of biological synapses such as paired-pulse facilitation (PPF, Supplementary Fig. 7) and spike rate-dependent plasticity (SRDP, Supplementary Fig. 8) based on the proposed mechanisms. Furthermore, the UDST exhibited consistent trends in PSC modulation with further increased number of pulses during SNDP demonstration (Supplementary Fig. 9), validating its reliability in synaptic dynamics emulation.

Neuromodulation property of the UDST

The dynamically reconfigurable synaptic properties of the optimized UDST, consisting of a 70 nm-thick pHID3 HkCTL and a 5 nm-thick pV3D3 TDL (UDST-3), were investigated by progressively increasing the magnitude of V_{PRE} by 1 V (pulse width = 50 ms, number of pulses = 100) (Fig. 4a). Three distinct operational schemes could be classified in the UDST according to the magnitude of V_{PRE} : potentiation (-14 V < V_{PRE} < -16 V), adaptation (-16.5 V < V_{PRE} < -19.5 V), and depression (-20 V < V_{PRE} < -22 V). In the potentiation region, the UDST exhibited positive conductance changes as the number of pulses increased. With the magnitude of the V_{PRE} increased, a larger number of -OH groups inside the pHID3 HkCTL aligned along with the applied field direction, leading to increased hole accumulation at the semiconductor/dielectric interface and enlarged positive conductance change. In the adaptation region, the conductance of the UDST initially increased with repeated V_{PRE} and saturated at a transition point, followed by conductance decrease as the pulse number increases, indicating the competitive interaction between hole accumulation via dipole polarization and hole depletion via charge trapping at the semiconductor/dielectric interface. In the depression region, the UDST showed negative conductance changes from the initial application of pulses. Here, charge trapping became dominant and progressively intensified as the magnitude of V_{PRE} increased, resulting in a significant decrease in channel conductance. Based on this analysis, we constructed a contour plot depicting the relationship between V_{PRE} , the number of applied pulses, and corresponding conductance change where blue regions, red regions, and white regions represent positive, negative, and zero conductance change, respectively (Fig. 4b). The contour plot clearly indicated that the transition from potentiation to adaptation occurred at V_{PRE} of -16.5 V, while the transition from adaptation to depression occurred at V_{PRE} of -19.5 V, with both potentiation and depression observed between these two thresholds depending on the number of applied pulses. Notably, this intermediate (adaptation) region reflects a history-dependent competitive

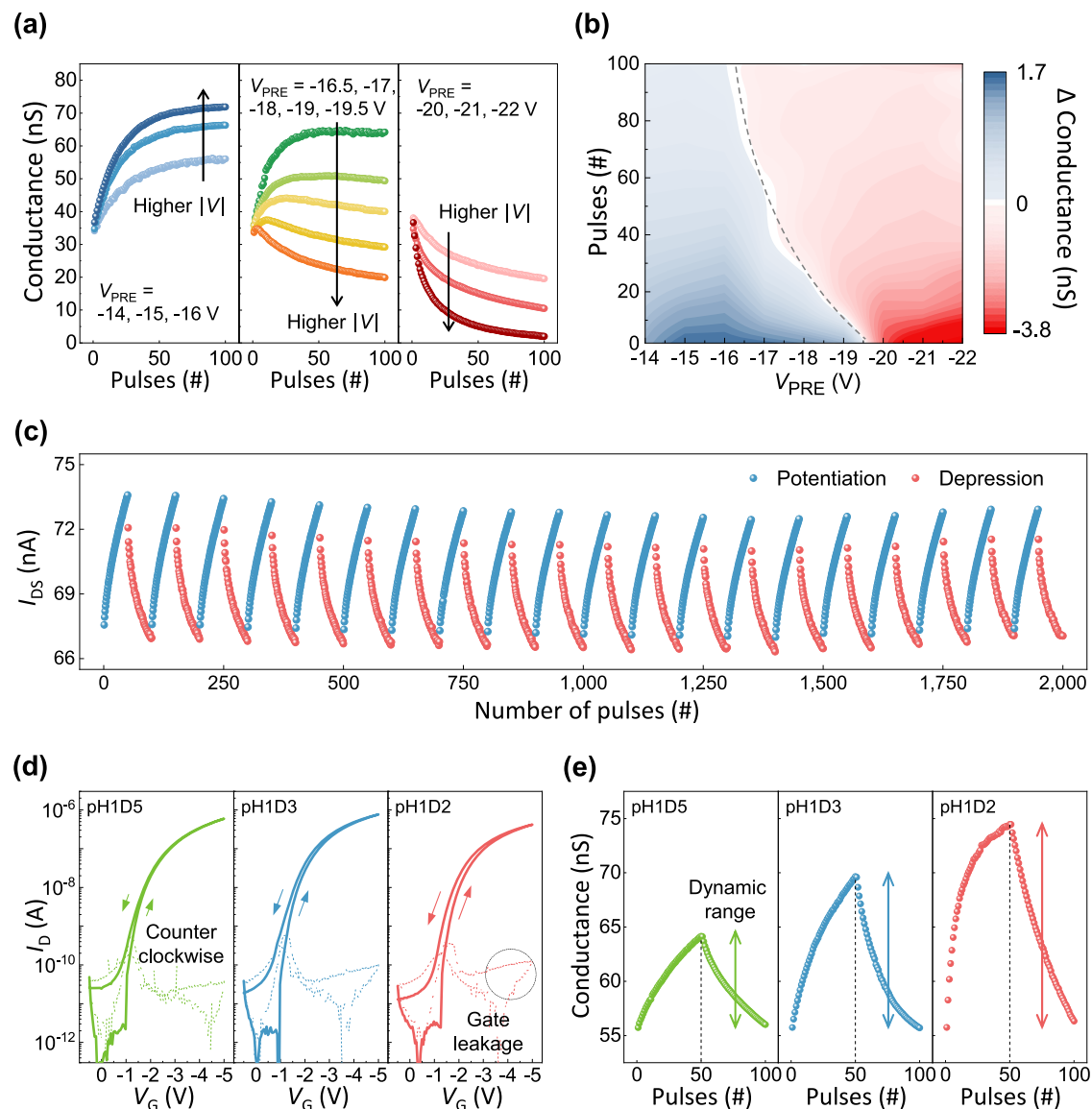


Fig. 4 | Neuromodulation property of the UDST. **a** Synapse characteristics of the UDST with progressive increase in V_{PRE} magnitude and **(b)** corresponding contour plot illustrating the relationship between V_{PRE} , pulse number, and conductance change. **c** Cycle endurance characteristics of the UDST with $V_{PRE} = -12$ V (blue dot) and $V_{PRE} = -20$ V (red dot). **d** Counter-clockwise hysteresis of the UDST with pH1D5, pH1D3, and pH1D2 HkCTL. The dashed line indicates gate leakage current of the

UDST. **e** Dynamic range of channel conductance for the UDST with pH1D5, pH1D3, and pH1D2 HkCTL. Synaptic characteristics of UDSTs with different HEA compositions were measured by applying the same conditions of potentiation/depression pulse set. All devices share the same configuration with a 5 nm-thick pV3D3 TDL and a 70 nm-thick p(HEA-co-DEGDVE) HkCTL, varying only in HEA composition.

interaction between dipole polarization and charge trapping. We found the thresholds of V_{PRE} for the transitions from potentiation to adaptation and from adaptation to depression were consistently observed across multiple devices, confirming the reliability and well-defined nature of the transition boundaries (Supplementary Fig. 10). These results clearly demonstrate that the conductance of the UDST can be dynamically reconfigured in response to V_{PRE} without polarity change, suggesting its potential applicability for neuromodulation of excitatory and inhibitory synapses in the visual system in response to incident light of arbitrary unipolar intensities.

The endurance of the potentiation/depression reconfiguration was also investigated by applying repetitive potentiation pulses (pulse voltage = -12 V, pulse width = 10 ms, number of pulses for each cycle = 50) and depression pulses (pulse voltage = -20 V, pulse width = 0.3 ms, number of pulses for each cycle = 50) to the UDST (Fig. 4c). The dynamic range of the UDST decreased less than 0.9% during 2000

pulses of repetitive potentiation and depression cycles. Also, the UDST maintained well-balanced plasticity updates, with a minimal conductance variation of only 0.3% between potentiation and depression, indicating the excellent electrical stability of the UDST. To the best of our knowledge, this is the first demonstration of reconfigurable artificial synapses that exhibits both potentiation/depression characteristics via unipolar input pulses with well-balanced plasticity updates coupled with superior durability (Supplementary Table 1).

In the human retinal system, the density of photoreceptors as well as the number of synapses formed between photoreceptors and bipolar cells and horizontal cells vary depending on the location within the retina⁴⁹. The UDST can reflect this density variation of synapses through the modulation of HEA composition in the p(HEA-co-DEGDVE) HkCTL, thereby mimicking a system where the magnitude of the signal transmitted to the brain varies based on the retinal location in response to the same visual stimuli. To examine the influence of HEA

composition in the p(HEA-co-DEGDVE) HkCTL on the synaptic properties of the UDST, transfer characteristics were measured for UDSTs incorporating pHID5, pHID3, and pHID2 (Fig. 4d). Counter-clockwise hysteresis was observed commonly from all the UDSTs, with the magnitude of hysteresis increasing in the order of UDSTs with pHID5, pHID3, and pHID2. Given the low operating voltage range, this hysteresis is attributed to the dipole polarization of the polar -OH groups in the p(HEA-co-DEGDVE) HkCTL, which correlates closely with the increasing amounts HEA composition. The synaptic characteristics of the UDST with pHID5, pHID3, and pHID2 were also assessed by applying 50 consecutive V_{PRE} for potentiation (-12 V for 50 ms) and depression (-20 V for 5 ms) (Fig. 4e). As the HEA composition in the p(HEA-co-DEGDVE) HkCTL increased, the dynamic range of channel conductance of the UDST increased from 10 nS (UDST with pHID5) to 20 nS (UDST with pHID2), demonstrating that the modulation of HEA composition in the p(HEA-co-DEGDVE) HkCTL enables systematic control of the dynamic range⁵⁰.

Implementation of artificial vision system

Based on the dynamically reconfigurable synaptic characteristics of the UDST, a self-adaptive artificial vision system was developed. The system included a CdS sensor with interfacial components and the optimized UDST with the 70 nm-thick pHID3 HkCTL and 5 nm-thick pV3D3 TDL, serving as the roles of photoreceptors, axons, and synapses, respectively (Fig. 5a). Note that the supporting circuit shown in Fig. 5a was designed solely to provide the necessary voltage conditions for operating the UDST and interfacing it with the sensor. Specifically, the system integrated a CdS sensor for electrical signal generation, amplification and voltage division stages for the generation of the drain and gate offset voltages, and a shunt resistor for PSC acquisition and utilization, collectively facilitating the continuous processing of external light signals. When voltage pulses of a fixed amplitude were applied through node 1, voltage division occurred between the CdS sensor and the series-connected resistor. The resistance of the CdS sensor dynamically adjusted to the light intensity, enabling light-induced amplitude modulation at node 2. This modulated signal underwent two stages of amplification to meet the voltage range requirements for PSC modulation in the UDST, with offset adjustments applied at node 3. The resulting amplified signal drove the UDST gate, inducing PSC changes at node 4, which can be measured through a shunt resistor connected to the source.

The PSC responses were measured at each node of the system under varying light conditions, where the light intensity was modulated from very bright (over 100 lux) to dim (under 10 lux) and then back to bright (Fig. 5b). The pulse width of all input signals was fixed at 150 ms to ensure that the system response was influenced solely by light intensity. The gate input voltage was successfully amplified to specific levels (-12 V for potentiation, -20 V for depression, and -16 V for adaptation), matching well with the operating voltage range of the UDST. Consequently, the PSC at node 4 decreased under -20 V (IPSC), increased under -12 V (EPSC), and remained nearly unchanged under -16 V, demonstrating dynamically reconfigurable synaptic characteristics for artificial vision systems. The PSC responses were further investigated under the range of light intensities that generated gate input voltages to induce potentiation and depression (Fig. 5c). When light intensities corresponding to gate input voltages of -10 V, -11 V, and -12 V were applied, EPSC was consistently obtained at node 4, where the magnitude of EPSC increased as the light intensities were increased. On the other hand, for light intensities corresponding to gate input voltages of -18 V, -19 V, and -20 V, IPSC was observed at node 4, showing increasingly pronounced changes with higher light intensities.

As discussed above, the biological visual system achieves scotopic and photopic adaptation by coordinating the functions of rod cells and cone cells to adjust light sensitivity. Similarly, in this artificial vision

system, PSC generation in response to varying light intensities was achieved across both dim and bright conditions, demonstrating the ability of the system to replicate the functions of both photoreceptors within a single UDST without polarity change. Moreover, the UDST exhibited rapid PSC responses to brief input signals (within 10 ms), enabling significantly faster information processing compared to biological systems (approximately 150 ms)^{51–53}. These findings highlight the potential of the UDST to provide advanced neuromodulation capabilities for artificial vision systems. As a proof of concept, PSC modulation under practical scenarios was evaluated by developing 3×3 UDST array (D1 to D9), whose device-to-device uniformity was confirmed by tightly overlapping transfer curves and a narrow ΔV_{TH} distribution (0.157 ± 0.02 V) (Supplementary Fig. 11). The PSC at each device in the array was monitored over time as a light-emitting source moved diagonally from the upper-left to the lower-right corner, followed by a transition to an environment with a significant increase in overall brightness (Fig. 5d). Here, the extracted PSC during each signal input sequence is also visualized as 3×3 pixel images, representing the PSC values ranging from 20 nA to 50 nA (Fig. 5e). As the source entered D1 region in a dim environment, a gate input voltage of -12 V was applied to D1, while adjacent devices D2 and D4 received -10 V due to the radial light distribution, resulting in a significant increase in the PSC of the pixel at the upper-left corner from 38.5 nA to 47.6 nA ($t = 1.1$ s). As the source moved to D5 region, a gate input voltage of -12 V was applied to D5, while -10 V was applied to adjacent devices D2, D4, D6, and D8, leading to the most substantial PSC increase from 38.0 nA to 48.9 nA at D5 ($t = 2.1$ s). Notably, the intensity at D1, previously the highest, gradually decreased, reflecting the new position of the source. Similarly, when the source moved to D9 region, the PSC at D9 increased to 45.7 nA, while PSC at D5 decreased to 39.8 nA ($t = 3.5$ s). Subsequently, as the surrounding environment became brighter, a gate input voltage exceeding -20 V was applied to all devices in the array. Since the source is located at D9 region, it was exposed to the brightest light among all devices due to the scattering light from the surrounding environment and the light reflected from the source. Consequently, the highest input gate voltage of -23 V was applied to D9. As a result, PSC across the array generally decreased, temporarily making the source location indistinguishable ($t = 3.7$ s). However, as the gate input voltage persisted, PSC at the D9 region exhibited the most significant decrease from 45.7 nA to 22.6 nA, enabling the identification of the source location even under the changed environmental conditions ($t = 4.0$ s). This response indicated an adjustment in light sensitivity to adapt to the new lightening condition, specifically photopic adaptation. Then, the environment transitioned to moderate lighting conditions. In this case, all devices in the array received a gate input voltage of -16 V, but the PSC remained almost unchanged, maintaining a certain level ($t = 5.9$ s). Considering the visual information processing performance, an artificial vision system must continuously adjust light sensitivity to environmental changes. Since the PSC remained nearly unchanged without saturating at an upper or lower limit due to potentiation or depression, this characteristic is highly advantageous for adapting to subsequent environmental transitions. Overall, the artificial vision system based on UDST array effectively processes visual data and transmits electrical signals under both dim and bright conditions. Initially, under the dim environment, the gradual increase in PSC, corresponding to scotopic adaptation, occurred in the UDST region where the light-emitting source was located. This was triggered by the projection of low-intensity light, allowing the system to identify the source location. Subsequently, under the bright environment, high-intensity light was projected, causing the overall PSC decrease of the UDST array. The most significant decrease occurred at the source's location as photopic adaptation, enabling object identification as well. Additionally, under the moderate lighting environment, the PSC practically remained unchanged. This characteristic allowed the UDST to operate

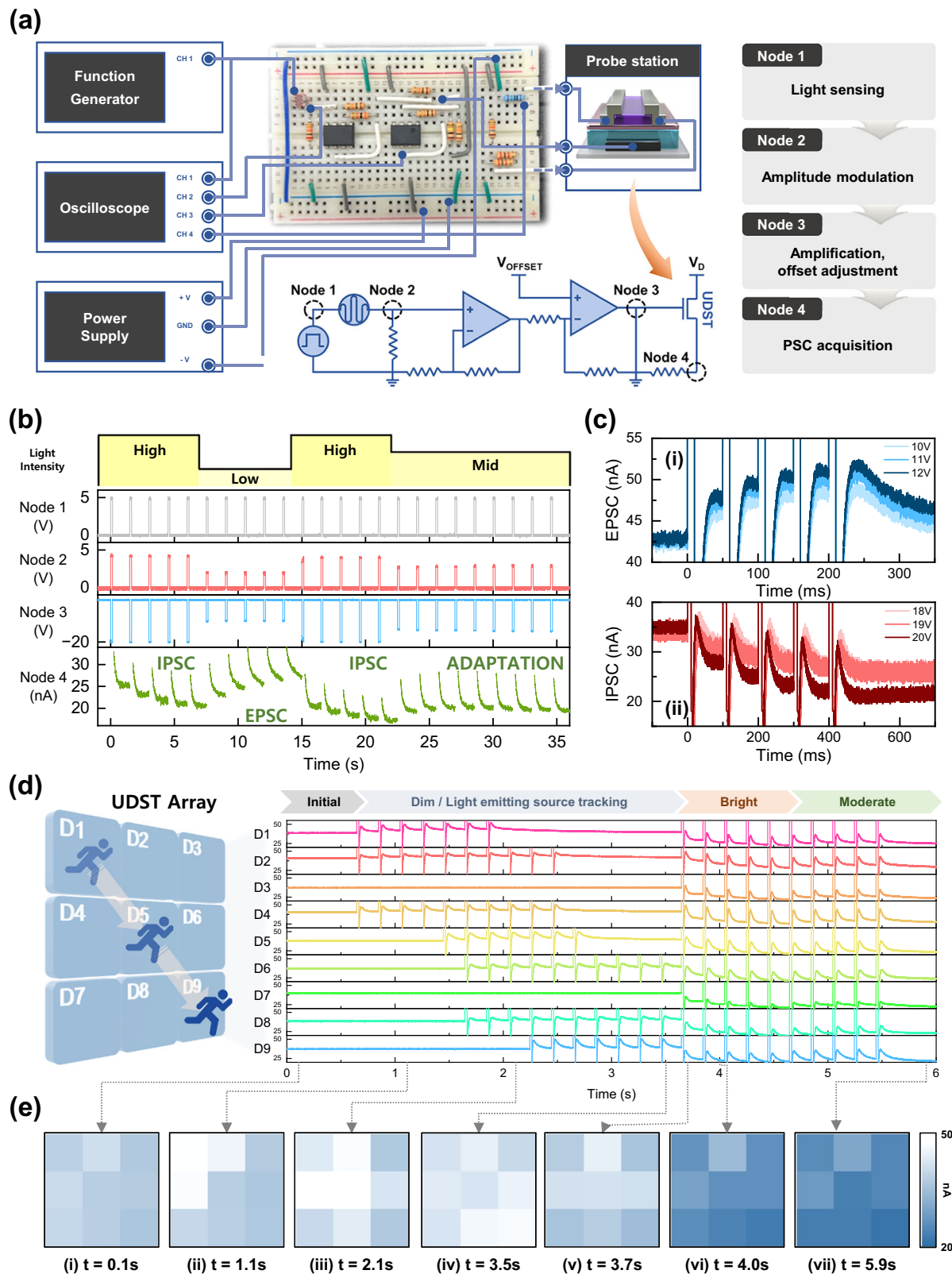


Fig. 5 | Implementation of artificial vision system. **a** Designed circuit schematic for artificial vision system including proper counterparts corresponding to photoreceptors and synapses. **b** Measurement results of PSC at each node according to light intensity. From top to bottom: input pulse (Node 1), pulse with amplitude proportional to light intensity due to voltage division (Node 2), amplified pulse including gate offset voltage (Node 3), and PSC obtained at the shunt resistance

(Node 4). **c** (i) EPSC trend correspond to the amplitude of gate input voltage. (ii) IPSC trend correspond to the amplitude of gate input voltage. **d** Schematic illustration of a light emitting source moving diagonally from the D1 region to the D9 region (left) and acquired PSC at each UDST within 3×3 array. **e** Visualized PSC extracted during each signal input sequence.

advantageously in adapting itself to subsequent environmental changes. This mirrored intensity-dependent sensory preprocessing of unipolar stimuli, achieved through synaptic interactions within the PNS. Especially, it is related to photoreceptors and afferent neurons in the human retinal system, where photoreceptor sensitivity is tuned to light intensity.

Discussion

In summary, we demonstrated a UDST as a fundamental building block for a self-adaptive artificial sensory system. By leveraging a bilayer gate dielectric structure, composed of p(HEA-co-DEGDVE) HkCTL and pV3D3 TDL, we achieved intensity-dependent synaptic plasticity, enabling potentiation, depression, and adaptation without requiring polarity inversion or external biasing elements. This design effectively mimics the unipolar synaptic dynamics of the PNS, offering a biologically inspired approach to sensory preprocessing. The systematic control of the thickness of the bilayer gate dielectric and chemical composition of the p(HEA-co-DEGDVE) HkCTL was realized, allowing tunable potentiation and depression characteristics of the UDST. By optimizing the device configuration, the UDST exhibited well-balanced plasticity updates and outstanding endurance, maintaining consistent synaptic properties over 2000 pulses of potentiation and depression. Furthermore, we integrated a 3×3 UDST array with a CdS sensor to realize an environment-adaptable artificial vision system that dynamically adjusts its response to illumination changes. By modulating the synaptic output currents based on both instantaneous and prior exposure conditions, the UDST array successfully demonstrated photopic and scotopic adaptation as well as real-time object tracking without the need for external computational circuits. Inspired by the preprocessing functions of the PNS, the unipolar operation of the UDST extends beyond visual signal processing, enabling its application across diverse sensory modalities. Its adaptive response allows for real-time sensitivity modulation based on stimulus intensity, establishing it as a versatile platform for self-regulating artificial sensory systems. We believe that the UDST, by closely emulating PNS-like preprocessing, marks a significant milestone in advancing artificial perception toward biological realism.

Methods

Materials & substrate

1,3,5-trimethyl-1,3,5-trivinyl cyclotrisiloxane (V3D3 monomer; 95%) was purchased from Gelest, and 2-hydroxyethyl acrylate (HEA monomer; >95%), diethylene glycol divinyl ether (DEGDVE monomer; 99%), and *tert*-butyl peroxide (TBPO; 97%) were purchased from Sigma-Aldrich for fabricating the ultrathin polymer dielectric layer. The pentacene, classified as a p-type semiconductor, was also purchased from Sigma-Aldrich as well. All the chemicals were used as received without additional purification. Glass substrate of 25×25 mm was cleaned with deionized (DI) water, acetone, and isopropyl alcohol sequentially for 20 min under ultrasonication, followed by blowing with dry N_2 gas.

Polymer dielectric deposition

Polymer dielectric layers were deposited by an iCVD system (iSAC Research). For poly(2-hydroxyethyl acrylate-co-diethylene glycol divinyl ether) [p(HEA-co-DEGDVE)] deposition, DEGDVE and TBPO were vaporized and injected into the iCVD chamber at the flow rates of 1.10 and 0.49 sccm, respectively. The flow rate of HEA monomer was also adjusted to 1.10, 0.66, and 0.22 sccm, respectively. Depending on the variation in the flow rate of the HEA monomer, p(HEA-co-DEGDVE) was named pHID2, pHID3, and pHID5, respectively. The substrate temperature and chamber pressure were maintained at 30 °C and 70 mTorr, respectively. The HEA, DEGDVE and TBPO were heated to 45, 50 and 30 °C, respectively to prompt their vaporization. For poly[1,3,5-trimethyl-1,3,5-trivinyl cyclotrisiloxane] (pV3D3) deposition, V3D3 and

TBPO were vaporized and injected into the iCVD chamber at flow rates of 4.16 and 1.47 standard cubic centimeter per minute (sccm), respectively. The substrate temperature and chamber pressure were maintained at 50 °C and 100 mTorr, respectively. The V3D3 and TBPO were heated to 40 and 30 °C, respectively to facilitate vaporization. The filament was heated to 130 °C to decompose the initiator into radicals for both processes.

Thin film characterization

The FT-IR spectra of the p(HEA-co-DEGDVE) were measured by an ALPHA FT-IR (Bruker Optics) with 64 scans in absorbance mode and the XPS spectra of the p(HEA-co-DEGDVE) were obtained by a sigma probe multipurpose spectrometer (Thermo VG Scientific). For UPS and REELS, the samples were loaded into the measuring equipment (Axis-Supra, Kratos) through the N_2 -filled pass glove box to avoid damage from moisture in ambient air. For EELS, 100 nm-thick films on a Si wafer were sliced using focused ion beam (FIB), then the cross-section of the prepared samples was analyzed with EELS built within a Cs-corrected scanning transmission electron microscope (Cs-STEM, ARM 200). UV-Vis spectra were obtained by the UV-vis spectrophotometer in absorbance mode (Lambda 1050, Perkin Elmer). AFM images of the pV3D3, pHEA and p(HEA-co-DEGDVE) were obtained using a scanning probe microscope (NX10, Park Systems). The thickness of the organic semiconductor and dielectric layers was measured using a spectroscopic ellipsometer (M2000, J. A. Woollam, USA).

Device fabrication

For gate electrodes, and metal-insulator-metal (MIM) devices, 50 nm-thick Al electrodes were thermally deposited at a deposition rate of $0.8 \text{ \AA s}^{-1}$. The polymer dielectric layer was deposited by the iCVD process, and the thickness was measured using an ellipsometry after the deposition. For the p-type semiconductor, pentacene was deposited at a deposition rate of $0.25 \text{ \AA s}^{-1}$. During the deposition, the thickness was monitored in-situ by quartz crystal microbalance (QCM). Then, WO_3 /Al electrodes (15 nm/50 nm) for sources/drain were deposited by thermal evaporation on the channel layer, in a vacuum under 3×10^{-6} Torr. The deposition rates of the WO_3 and Al were set to 0.3 and $0.8 \text{ \AA s}^{-1}$, respectively. The channel dimensions were 1000 μm for width and 200 μm for length. The area of the back-gate dielectric (corresponding to the overlapping region of the gate and dielectric) is $4 \times 10^6 \mu\text{m}^2$ (Supplementary Fig. 12).

Device characterization

The electrical properties of all the devices were measured using a B1500A semiconductor device analyzer (Agilent Technologies) in the N_2 -filled glove box at room temperature.

Circuit fabrication & characterization

In the artificial vision system circuit, DC voltage was generated using Keysight E3631A power supply. Also, the circuit components GL5537 for CdS sensor made by GEC and OPA445AP for OP-AMP made by Texas Instruments were used. All electrical characterization was performed using Tektronix AFG3102C function generator and Keysight DSO-X 3024 T oscilloscope.

Data availability

The data within the article and its supplementary Information are available from the corresponding authors upon request.

References

1. Jung, Y. H., Park, B., Kim, J. U. & Kim, T. i. Bioinspired electronics for artificial sensory systems. *Adv. Mater.* **31**, 1803637 (2019).
2. Sun, F., Lu, Q., Feng, S. & Zhang, T. Flexible artificial sensory systems based on neuromorphic devices. *ACS Nano* **15**, 3875–3899 (2021).

3. Heng, W., Solomon, S. & Gao, W. Flexible electronics and devices as human-machine interfaces for medical robotics. *Adv. Mater.* **34**, 2107902 (2022).
4. Seo, D.-G. et al. Versatile neuromorphic electronics by modulating synaptic decay of single organic synaptic transistor: From artificial neural networks to neuro-prosthetics. *Nano Energy* **65**, 104035 (2019).
5. Wang, J. et al. Artificial sense technology: emulating and extending biological senses. *ACS Nano* **15**, 18671–18678 (2021).
6. Choudhry, H. H., Lee, D. H., Bag, A. & Lee, N.-E. A flexible artificial chemosensory neuronal synapse based on chemoreceptive ionogel-gated electrochemical transistor. *Nat. Commun.* **14**, 821 (2023).
7. Kim, Y. et al. A bioinspired flexible organic artificial afferent nerve. *Science* **360**, 998–1003 (2018).
8. Qu, S. et al. An artificially-intelligent cornea with tactile sensation enables sensory expansion and interaction. *Nat. Commun.* **14**, 7181 (2023).
9. Jiang, C. et al. Mammalian-brain-inspired neuromorphic motion-cognition nerve achieves cross-modal perceptual enhancement. *Nat. Commun.* **14**, 1344 (2023).
10. Gong, J. et al. An artificial visual nerve for mimicking pupil reflex. *Matter* **5**, 1578–1589 (2022).
11. Demb, J. B. Functional circuitry of visual adaptation in the retina. *J. Physiol.* **586**, 4377–4384 (2008).
12. Lee, G. J., Choi, C., Kim, D. H. & Song, Y. M. Bioinspired artificial eyes: optic components, digital cameras, and visual prostheses. *Adv. Funct. Mater.* **28**, 1705202 (2018).
13. Berco, D. & Shenp Ang, D. Recent progress in synaptic devices paving the way toward an artificial cogni-retina for bionic and machine vision. *Adv. Intell. Syst.* **1**, 1900003 (2019).
14. Han, X. et al. Recent progress in optoelectronic synapses for artificial visual-perception system. *Small Struct.* **1**, 2000029 (2020).
15. Cho, S. W., Kwon, S. M., Kim, Y.-H. & Park, S. K. Recent progress in transistor-based optoelectronic synapses: from neuromorphic computing to artificial sensory system. *Adv. Intell. Syst.* **3**, 2000162 (2021).
16. Cho, S. W., Jo, C., Kim, Y.-H. & Park, S. K. Progress of materials and devices for neuromorphic vision sensors. *Nano-Micro Lett.* **14**, 203 (2022).
17. Euler, T., Haverkamp, S., Schubert, T. & Baden, T. Retinal bipolar cells: elementary building blocks of vision. *Nat. Rev. Neurosci.* **15**, 507–519 (2014).
18. Shi, L. et al. Flexible retinomorph vision sensors with scotopic and photopic adaptation for a fully flexible neuromorphic machine vision system. *SmartMat.* **5**, e1285 (2024).
19. Liao, F. et al. Bioinspired in-sensor visual adaptation for accurate perception. *Nat. Electron.* **5**, 84–91 (2022).
20. Graydon, C. W. et al. Synaptic transfer between rod and cone pathways mediated by All amacrine cells in the mouse retina. *Curr. Biol.* **28**, 2739–2751.e2733 (2018).
21. Hu, M. et al. A non-linear two-dimensional float gate transistor as a lateral inhibitory synapse for retinal early visual processing. *Mater. Horiz.* **9**, 2335–2344 (2022).
22. Yu, J. M. et al. All-solid-state ion synaptic transistor for Wafer-scale integration with electrolyte of a nanoscale thickness. *Adv. Funct. Mater.* **31**, 2010971 (2021).
23. Yu, J. M. et al. Bio-inspired organic synaptor with in situ ion-doped ultrathin polyelectrolyte containing acetylcholine-like cation. *Small* **20**, 2312283 (2024).
24. Wang, H., Chen, Y., Ni, Z. & Samorì, P. An electrochemical-electret coupled organic synapse with single-polarity driven reversible facilitation-to-depression switching. *Adv. Mater.* **34**, 2205945 (2022).
25. Lee, C., Choi, J., Lee, C., Yoo, H. & Im, S. G. Exploring new logic devices: Unlocking potential with floating-gate transistor. *Appl. Phys. Rev.* **11**, 031317 (2024).
26. Bu, Y. et al. Ferroelectrics-electret synergetic organic artificial synapses with single-polarity driven dynamic reconfigurable modulation. *Adv. Funct. Mater.* **33**, 2213741 (2023).
27. Li, M. et al. Tailoring neuroplasticity in a ferroelectric-gated multi-terminal synaptic transistor by bi-directional modulation for improved pattern edge recognition. *Adv. Funct. Mater.* **33**, 2307986 (2023).
28. Kwon, S. M. et al. Environment-adaptable artificial visual perception behaviors using a light-adjustable optoelectronic neuromorphic device array. *Adv. Mater.* **31**, 1906433 (2019).
29. Choi, J. et al. Spontaneous generation of a molecular thin hydrophobic skin layer on a sub-20 nm, high-k polymer dielectric for extremely stable organic thin-film transistor operation. *ACS Appl. Mater. Interfaces* **11**, 29113–29123 (2019).
30. Tsai, T. D., Chang, J. W., Wen, T. C. & Guo, T. F. Manipulating the Hysteresis in Poly (vinyl alcohol)-Dielectric Organic Field-Effect Transistors Toward Memory Elements. *Adv. Funct. Mater.* **23**, 4206–4214 (2013).
31. Hwang, D. K. et al. Top-gate organic field-effect transistors with high environmental and operational stability. *Adv. Mater.* **10**, 1293–1298 (2011).
32. Choi, J. et al. Flexible, low-power thin-film transistors made of vapor-phase synthesized high-k, ultrathin polymer gate dielectrics. *ACS Appl. Mater. Interfaces* **9**, 20808–20817 (2017).
33. Kang, J. et al. Ultrathin, solvent-resistant dielectric for monolithic fabrication of low-power, intrinsically stretchable active-matrix electronic skin. *Device* **2**, 100426 (2024).
34. Moon, H. et al. Synthesis of ultrathin polymer insulating layers by initiated chemical vapour deposition for low-power soft electronics. *Nat. Mater.* **14**, 628–635 (2015).
35. Pak, K., Seong, H., Choi, J., Hwang, W. S. & Im, S. G. Synthesis of ultrathin, homogeneous copolymer dielectrics to control the threshold voltage of organic thin-film transistors. *Adv. Funct. Mater.* **26**, 6574–6582 (2016).
36. Jang, S. et al. Vapor-phase deposited polymer dielectric layers for organic electronics: design, characteristics, and applications. *Korean J. Chem. Eng.* **42**, 1931 (2024).
37. Jeong, K. et al. A hyperelastic adhesive forming multiple neutral planes even at extreme temperatures. *Chem. Eng. J.* **480**, 148151 (2024).
38. Park, H., Yoo, H., Lee, C., Kim, J.-J. & Im, S. G. Multi-stage organic logic circuits using via-hole-less metal interconnects. *IEEE Electron Device Lett.* **41**, 1685–1687 (2020).
39. Choi, J. et al. Vertically stacked, low-voltage organic ternary logic circuits including nonvolatile floating-gate memory transistors. *Nat. Commun.* **13**, 2305 (2022).
40. Lin-Vien, D., Colthup, N. B., Fateley, W. G. & Grasselli, J. G. The handbook of infrared and Raman characteristic frequencies of organic molecules (Elsevier, 1991).
41. Sharp, J. et al. Calcium thin film growth on a cyano-substituted poly (p-phenylene vinylene): Interface structure and energetics. *J. Phys. Chem. C* **117**, 23781–23789 (2013).
42. Lee, C. et al. Systematic control of negative transconductance in organic heterojunction transistor for high-performance, low-power flexible ternary logic circuits. *Small* **17**, 2103365 (2021).
43. Choe, Y.-S. et al. Crosslinked polymer-mixture gate insulator for high-performance organic thin-film transistors. *Org. Electron.* **36**, 171–176 (2016).
44. Han, W., Yoshida, H., Ueno, N. & Kera, S. Electron affinity of penta-cene thin film studied by radiation-damage free inverse photo-emission spectroscopy. *Appl. Phys. Lett.* **103**, 123303 (2013).

45. Pak, K., Choi, J., Lee, C. & Im, S. G. Low-power, flexible nonvolatile organic transistor memory based on an ultrathin bilayer dielectric stack. *Adv. Electron. Mater.* **5**, 1800799 (2019).
46. Hwang, D., Oh, M. S., Hwang, J. M., Kim, J. H. & Im, S. Hysteresis mechanisms of pentacene thin-film transistors with polymer/oxide bilayer gate dielectrics. *Appl. Phys. Lett.* **92**, 013304 (2008).
47. Fan, J. et al. Performance enhancement of pentacene-based organic field-effect transistor by inserting a WO₃ buffer layer. *J. Semiconductors* **35**, 064004 (2014).
48. Toprasertpong, K., Takenaka, M. & Takagi, S. On the strong coupling of polarization and charge trapping in HfO₂/Si-based ferroelectric field-effect transistors: overview of device operation and reliability. *Appl. Phys. A* **128**, 1114 (2022).
49. Wässle, H. & Riemann, H. The mosaic of nerve cells in the mammalian retina. *Proc. R. Soc. Lond. Ser. B. Biol. Sci.* **200**, 441–461 (1978).
50. Shen, C.-K., Chaurasiya, R., Chen, K.-T. & Chen, J.-S. Synaptic emulation via ferroelectric P (VDF-TrFE) reinforced charge trapping/detrapping in Zinc-Tin oxide transistor. *ACS Appl. Mater. Interfaces* **14**, 16939–16948 (2022).
51. Hegd , J. Time course of visual perception: coarse-to-fine processing and beyond. *Prog. Neurobiol.* **84**, 405–439 (2008).
52. VanRullen, R. & Thorpe, S. J. The time course of visual processing: from early perception to decision-making. *J. Cogn. Neurosci.* **13**, 454–461 (2001).
53. Potter, M. C., Wyble, B., Hagmann, C. E. & McCourt, E. S. Detecting meaning in RSVP at 13 ms per picture. *Atten., Percept., Psychophys.* **76**, 270–279 (2014).

Acknowledgements

This work was supported by the Technology Innovation Program (1415181712 and RS-2022-00144300 to S.G.I.) funded by the Ministry of Trade, Industry & Energy (MOTIE, Korea). This work was supported by Samsung Electronics Co., Ltd (IO201211-08088-01 to S.G.I.). This work was supported by Korea Institute for Advancement of Technology (KIAT) grant funded by the Korea Government(MOTIE) (RS-2023-KI002692, HRD Program for Industrial Innovation to S.G.I.). This work was also supported by National Research Foundation of Korea (NRF) and the Korea Basic Science Institute funded by the Ministry of Science and ICT (RS-2024-00406418 and RS-2024-00403917 to J.H.Y.).

Author contributions

S.J., K.S., CR.L., J.H.Y. and S.G.I. conceived the idea and designed the experiments. S.J., K.S. and CR.L. designed, fabricated, and measured all

the devices. T.N., M.J., J.P., J.C., CH.L. assisted with device fabrication and characterization. S.J., K.S., CR.L., J.H.Y. and S.G.I. wrote the manuscript. All authors reviewed the manuscript and discussed the results. S.J., K.S. and CR.L. contributed equally to this work. J.H.Y. and S.G.I. contributed equally as corresponding authors.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41467-025-63073-2>.

Correspondence and requests for materials should be addressed to Jung Ho Yoon or Sung Gap Im.

Peer review information *Nature Communications* thanks Wentao Xu, Jianhua Yang and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. A peer review file is available.

Reprints and permissions information is available at <http://www.nature.com/reprints>

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

  The Author(s) 2025